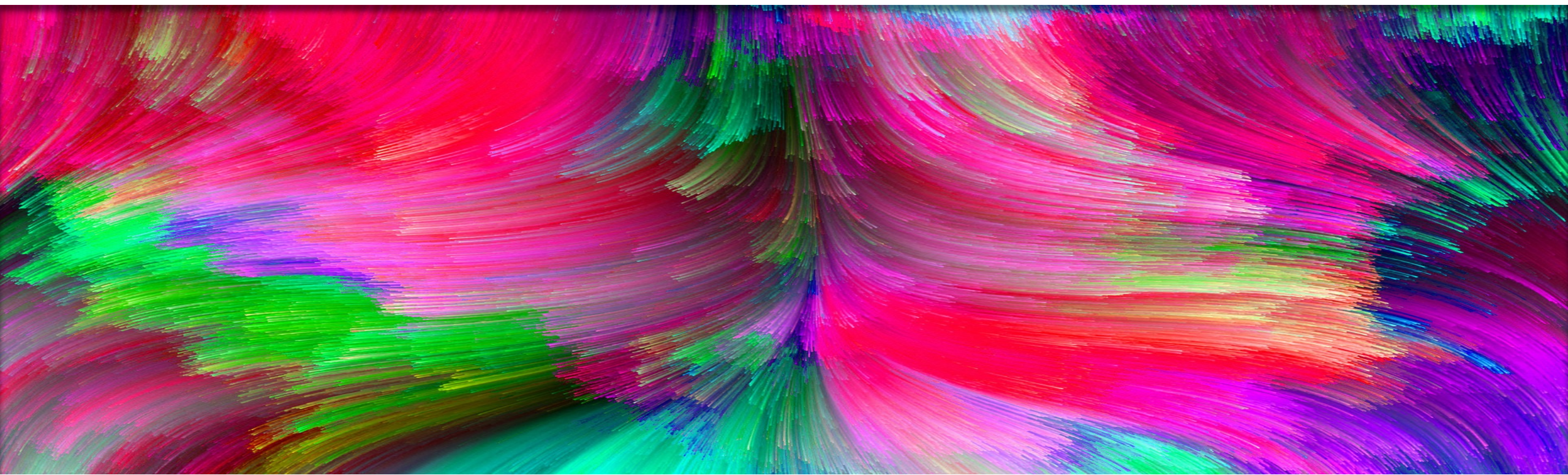




Artificial Intelligence
and Machine Learning

Universität St.Gallen

School of Computer Science (SCS)
Artificial Intelligence & Machine Learning [AI:ML]



Fall Term 26 - 7,854,1.00 MCS Machine Learning

Lab 02 – Coding Challenge Kick-Off

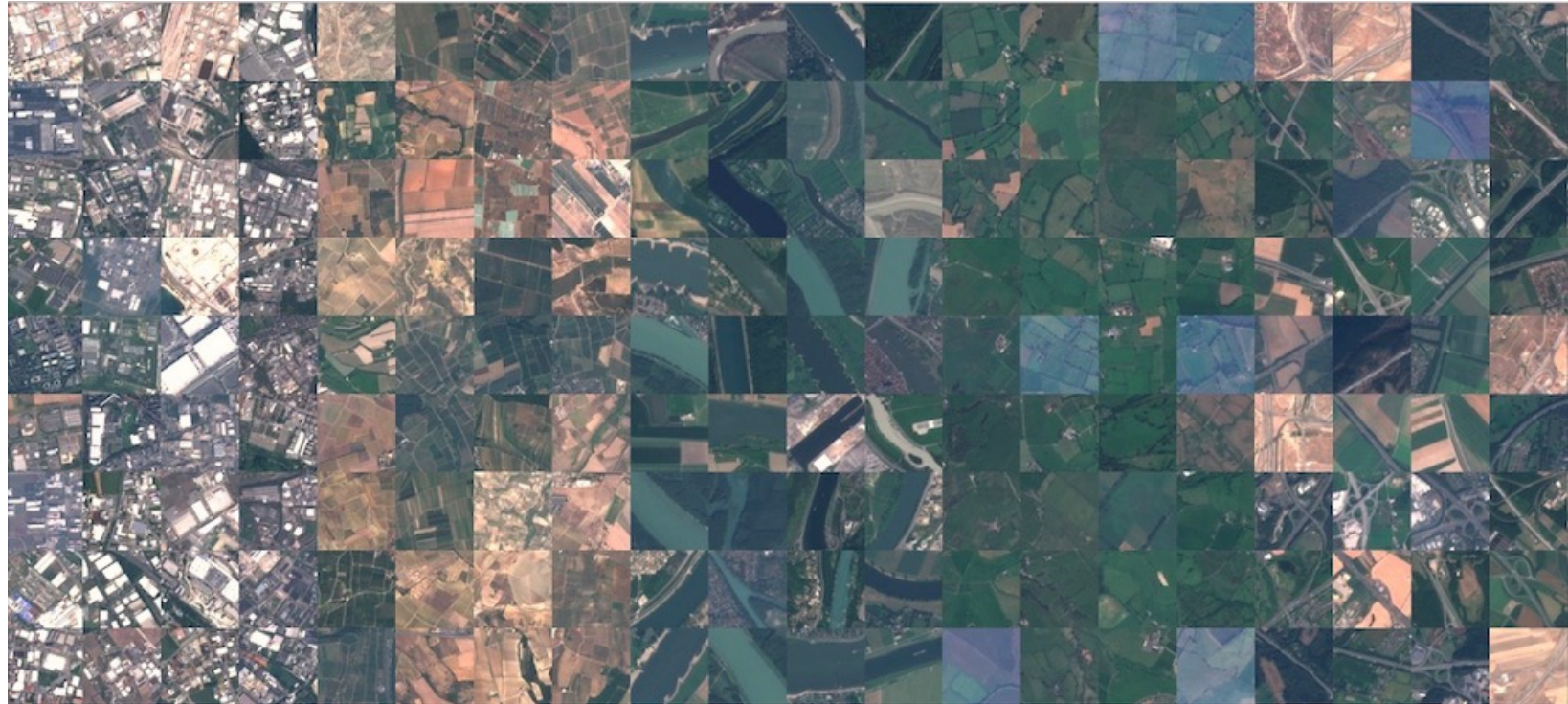
Aron, Léo

From insight to impact.

Coding Challenge

kaggle

Satellite Image Classification - Eurosat



Helber, P., Bischke, B., Dengel, A., & Borth, D. (2019). Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7), 2217-2226.

Course Logistics Updates

Weekly Updates and Questions

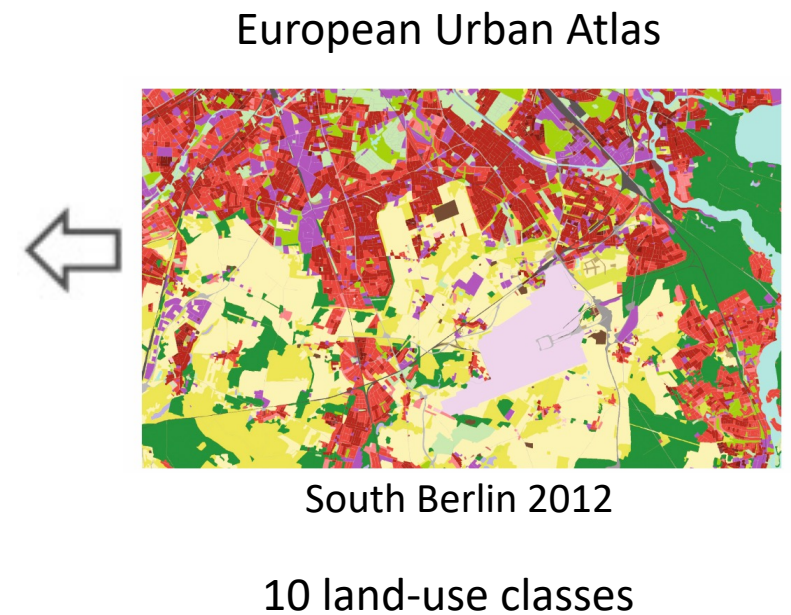
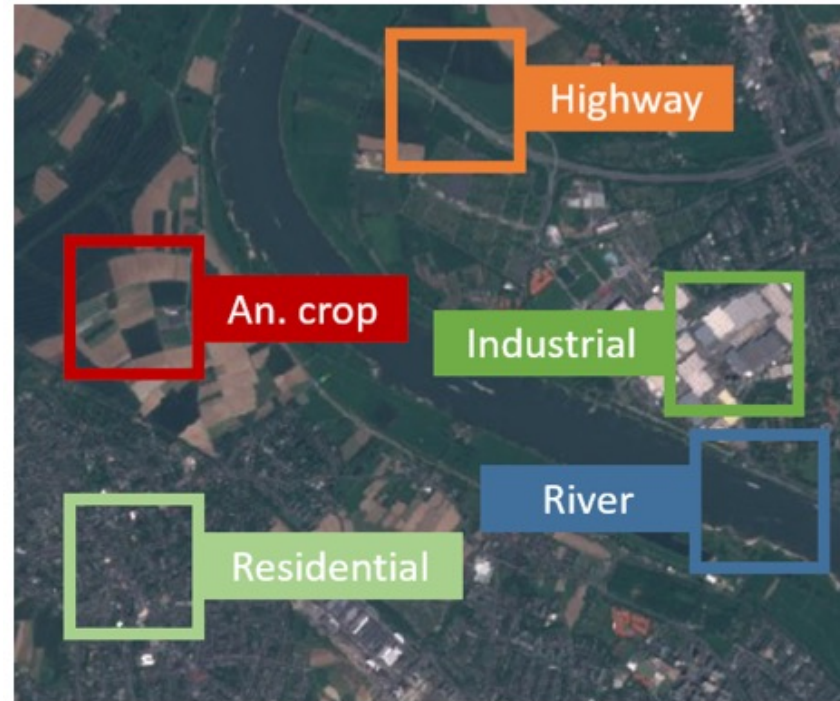
Course Labs and Challenge Roadmap

Event Type	Date	Description	Teaching Assistant	CC Phase
Lab 1	14 Sep 2026	Co-Lab Introduction	Léo/Aron	Introduction
CC 1	21 Sep 2026	Coding Challenge - Kick-Off	Léo/Aron	Ramp-Up Phase
Lab 2	28 Sep 2026	Support Vector Machine (SVM)	Léo/Aron	Ramp-Up Phase
Lab 3	05 Oct 2026	Fully Connected Neural Networks	Léo/Aron	Ramp-Up Phase
Lab 4	12 Oct 2026	Convolutional Neural Networks (CNNS)	Léo/Aron	Ramp-Up Phase
Lab 5	19 Oct 2026	Long short-term memory (LSTM)	Léo/Aron	Ramp-Up Phase
Semester Break				

EuroSAT



27,000 Sentinel-2 image patches



- Helber, P., Bischke, B., Dengel, A., & Borth, D. (2019). Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7), 2217-2226.
- Helber, P., Bischke, B., Dengel, A., & Borth, D. (2018, July). Introducing eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. In *IGARSS 2018-2018 IEEE international geoscience and remote sensing symposium* (pp. 204-207). IEEE.
- <https://land.copernicus.eu/local/urban-atlas>

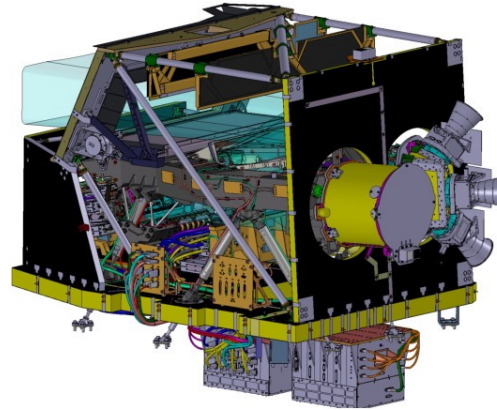
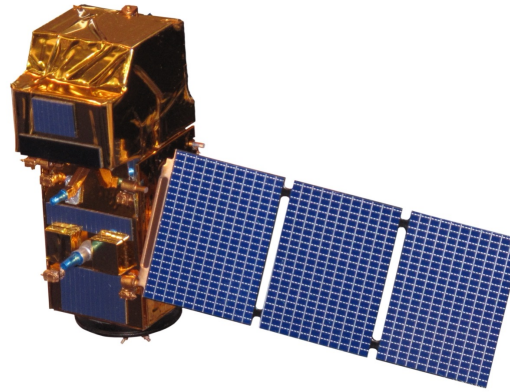


Sentinel-2

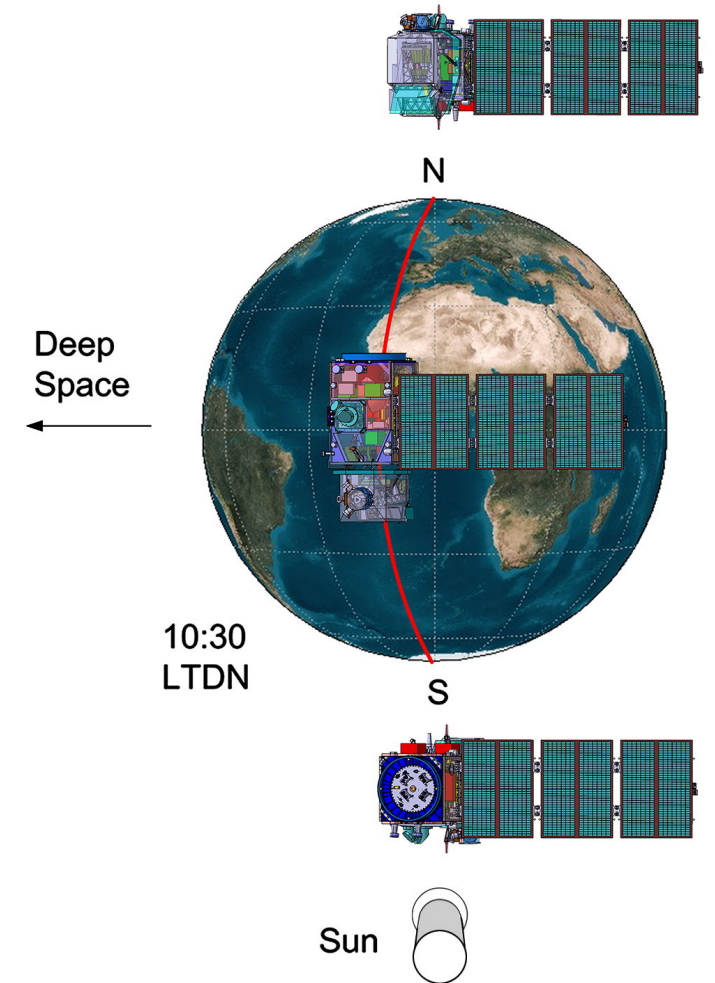
- ESA satellite mission with two satellites
- Launched 2015
- Sun-synchronous orbit
- 786km altitude
- 5-day revisit rate

Payload: Multi Spectral Instrument

- 290km field of view
- 13 spectral bands
- Visible, near- and short wave infra-red
- Spatial resolutions 10 to 60m



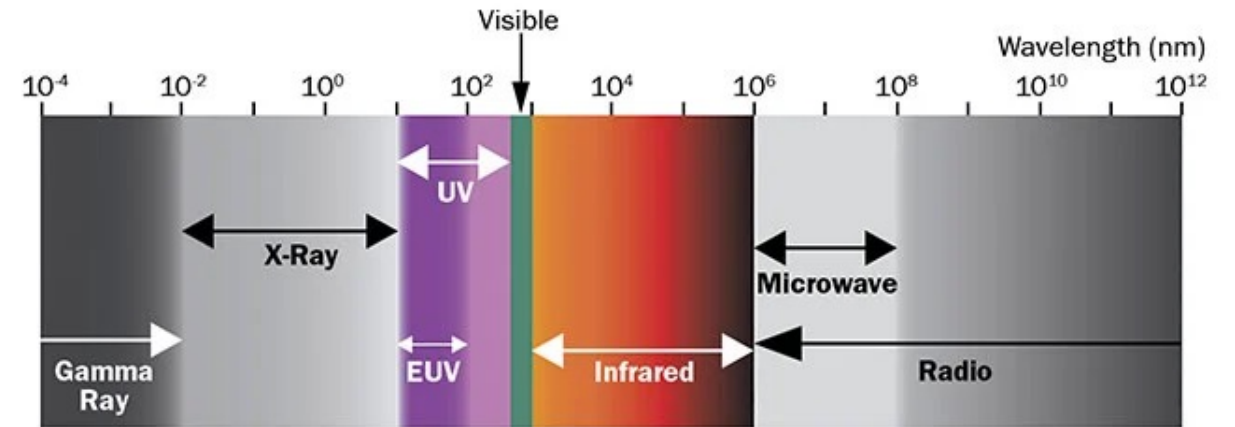
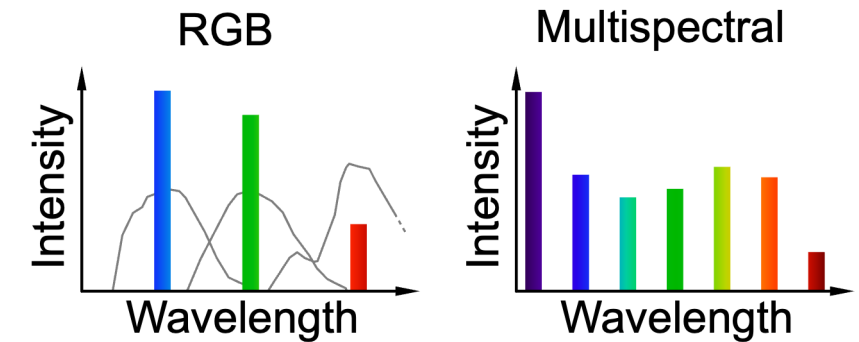
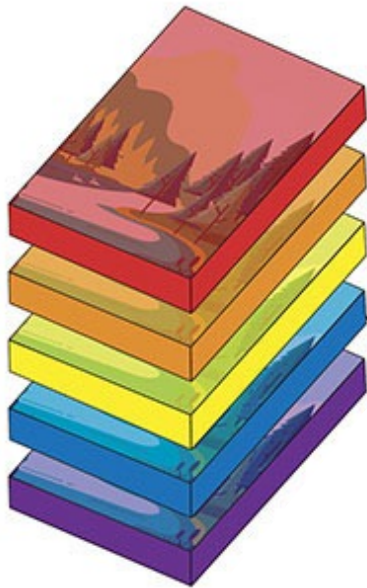
View on Equator



Multi Spectral Imaging

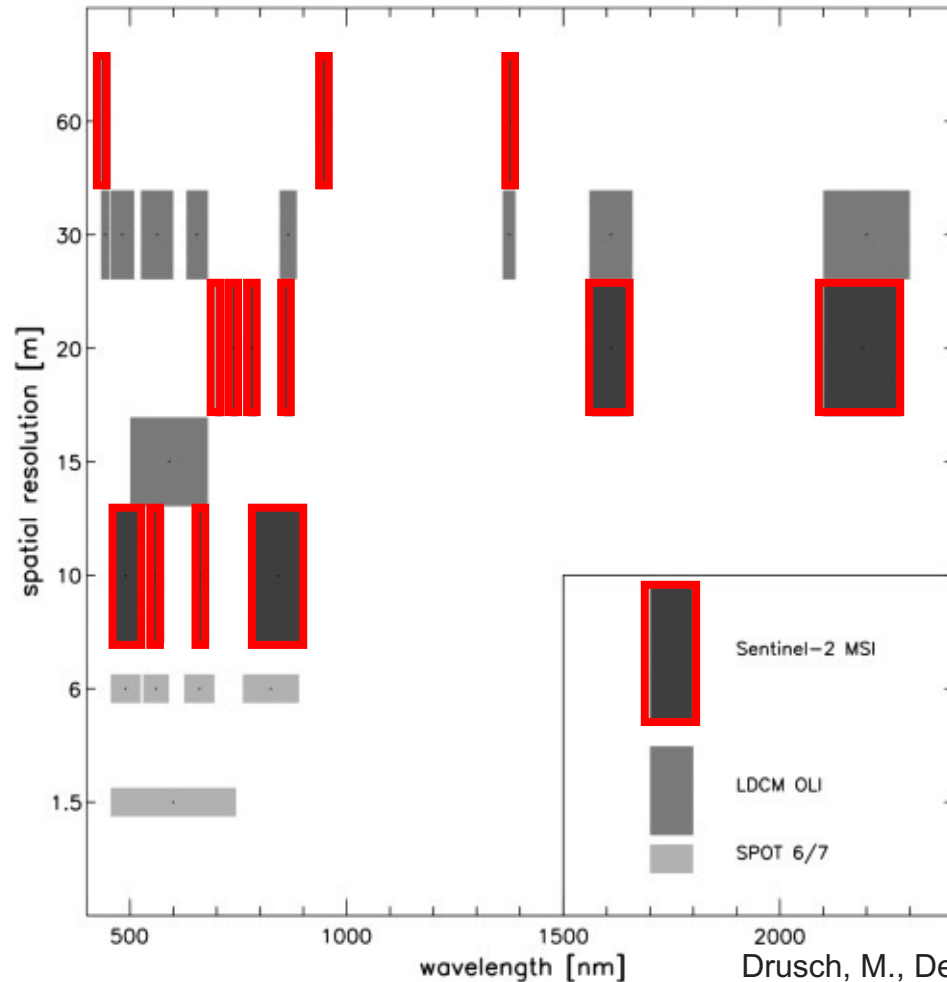
Measure electromagnetic radiation at wavelengths outside the visible spectrum

Multi-spectral image



- Drusch, M., Del Bello, U., Carlier, S., Colin, O., Fernandez, V., Gascon, F., ... & Bargellini, P. (2012). Sentinel-2: ESA's optical high-resolution mission for GMES operational services. *Remote sensing of Environment*, 120, 25-36.
- https://www.photonics.com/Articles/Hyperspectral_and_Multispectral_Imaging/a65595
- <https://wiki.tum.de/display/zfp/Hyper+and+multispectral+imaging>

Sentinel-2 Multi Spectral Imaging



13 spectral bands in the visible, near- and short wave infra-red

Spatial resolutions between 10 and 60m

Band	Spatial Resolution <i>m</i>	Central Wavelength <i>nm</i>
B01 - Aerosols	60	443
B02 - Blue	10	490
B03 - Green	10	560
B04 - Red	10	665
B05 - Red edge 1	20	705
B06 - Red edge 2	20	740
B07 - Red edge 3	20	783
B08 - NIR	10	842
B08A - Red edge 4	20	865
B09 - Water vapor	60	945
B10 - Cirrus	60	1375
B11 - SWIR 1	20	1610
B12 - SWIR 2	20	2190

Sentinel-2 Multi Spectral Imaging

B4, B3, B2



B11, B8, B2

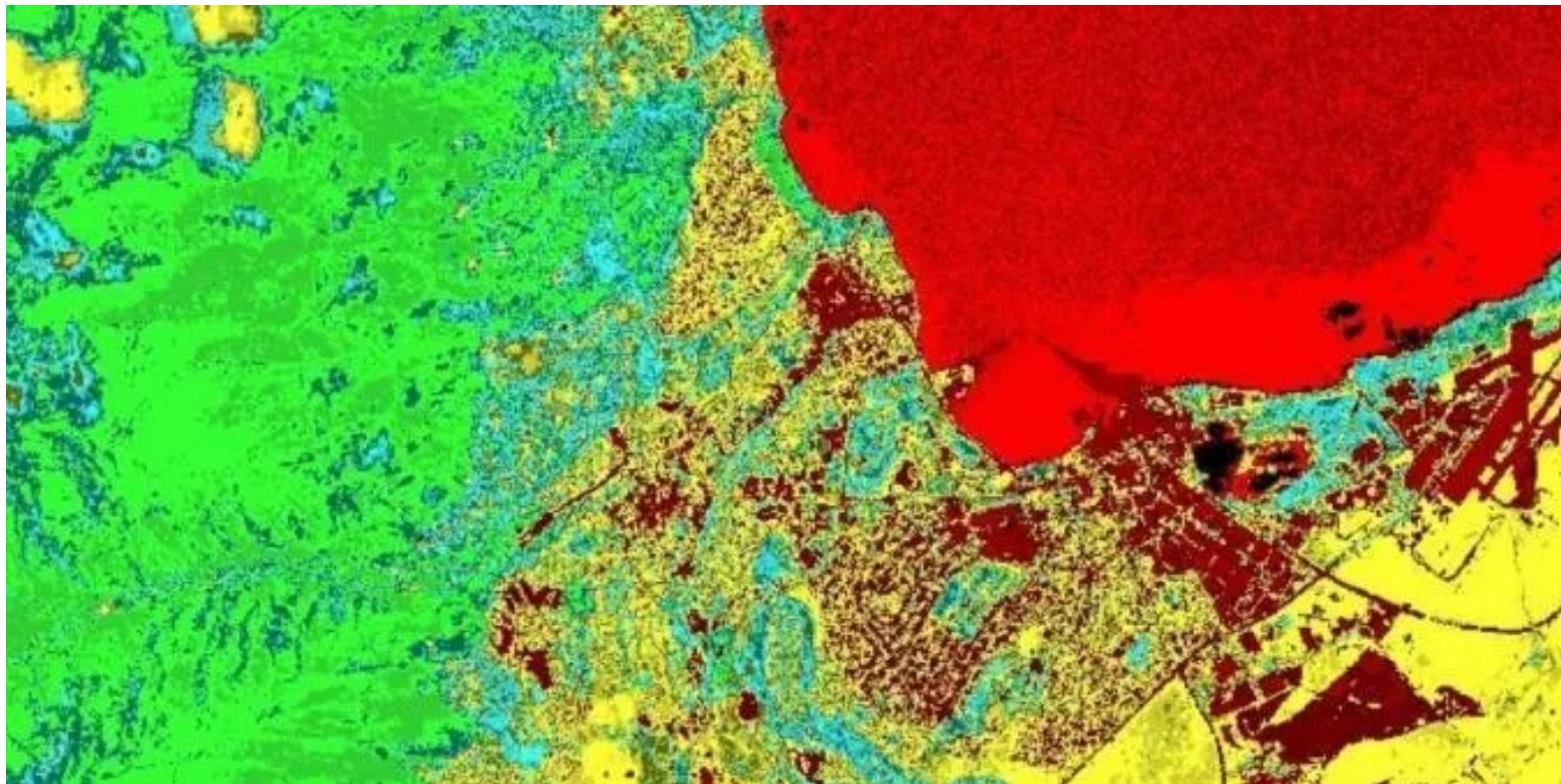


B12, B8A, B5



B8, B4, B3

Normalized difference vegetation index (NDVI):
 $(B8-B4)/(B8+B4)$



Sentinel-2 Products

Level-1C

Top of Atmosphere
reflectance



Level-2A

Bottom of Atmosphere
reflectance

Cirrus cloud mask

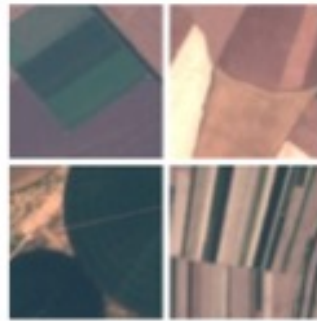
Eurosat Classes



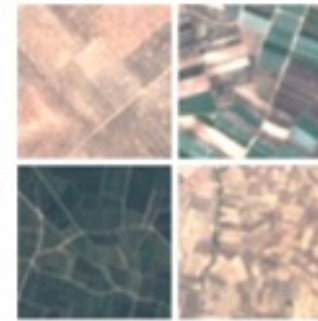
(a) Industrial Buildings



(b) Residential Buildings



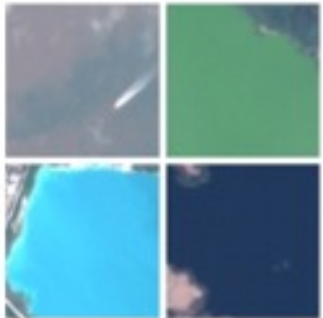
(c) Annual Crop



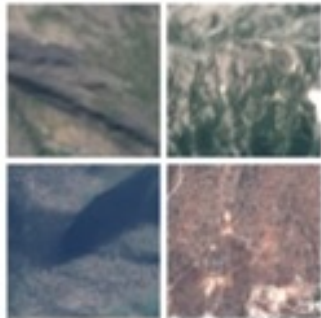
(d) Permanent Crop



(e) River



(f) Sea & Lake



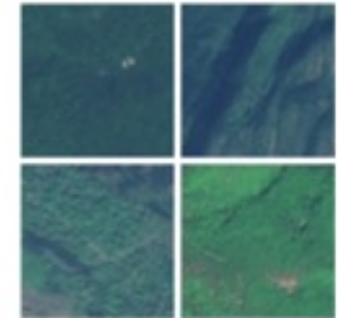
(g) Herbaceous Vegetation



(h) Highway



(i) Pasture



(j) Forest

27,000 13x64x64 pixel images (2,000 – 3,000 per class)

Helber, P., Bischke, B., Dengel, A., & Borth, D. (2019). Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7), 2217-2226.

Challenge: Land Cover Classification

Training Data

- Eurosat
- Level-1C
- 13 Bands
- GeoTiff



Test Data

- Atmospheric Correction
- Level-2A
- 12 bands
- Without B10 (Cirrus)
- npy files
- 4200 images



Kaggle Overview

- Platform for data science/machine learning challenges
- Challenge organizer provides data and task
- Participants can upload their solutions on test data
- Kaggle scores the solutions and creates a leaderboard
- Lots of additional functionality
 - Discussion board
 - Code sharing
 - Data hosting
 - ...

Competitions

Grow your data science skills by competing in our exciting competitions. Find help in the [documentation](#) or learn about [Community Competitions](#).

[Host a Competition](#)[Your Work](#)[Filters](#)[All competitions](#)[Entered](#)[Hosted](#)[Featured](#)[Research](#)[Getting Started](#)[Playground](#)[Analytics](#)[Community](#)

Active Competitions

[Hotness](#)

Ubiquant Market Prediction

Make predictions against future market d...

Featured

Code Competition · 2174 Teams

\$100,000 a month to go

NBME - Score Clinical Patient Notes

Identify Key Phrases in Patient Notes fro...

Featured

Code Competition · 775 Teams

\$50,000 a month to go

H&M Personalized Fashion Recommendations

Provide product recommendations based ...

Featured

1357 Teams

\$50,000 2 months to go

Happywhale - Whale and Dolphin Identification

Identify whales and dolphins by unique ch...

Research

1132 Teams

\$25,000 a month to go

BirdCLEF 2022

Identify bird calls in soundscapes

Research

Code Competition · 242 Teams

\$10,000 2 months to go

Tabular Playground Series - Mar 2022

Practice your ML skills on this approacha...

Playground

721 Teams

Swag 11 days to go

Kore 2022 - Beta

Collect the maximum amount of Kore agai...

Playground

Simulation Competition · 21 Teams

Swag 11 days to go

GeoLifeCLEF 2022 - LifeCLEF 2022 x FGVC9

Location-based species presence predicti...

Research

20 Teams

Knowledge 2 months to go

www.kaggle.com

AIML Coding Challenge

- You need a Kaggle account to compete in the challenge
- Teams of three people:
 - Create teams on Kaggle in the `Team` tab
- Use the Kaggle discussion board to discuss with other students
 - We do not monitor conversations and might not respond there

Join challenge at:

- <https://www.kaggle.com/t/22064d36e9bb49f98a6ac0e0e8dba97d>

Permanent challenge link:

- <https://www.kaggle.com/competitions/7-854-1-00-machine-learning-2026-coding-challenge/overview>

Kaggle Submission Workflow

1. Download EuroSat and testset
2. Develop a model on EuroSat data
3. Apply the model on the testset
4. Upload a CSV with testset predictions to Kaggle
 - Mapping of test sample ids to predicted class label

Note: Kaggle provides **public** and **private** leaderboards

The final ranking will be based on the private leaderboard

```
sample_submission.csv
test_id,label
0,HerbaceousVegetation
1,Residential
2,HerbaceousVegetation
3,SeaLake
4,PermanentCrop
5,River
6,PermanentCrop
7,Industrial
8,Residential
9,Residential
10,River
```

Performance metric

Kaggle scores the submissions based on **accuracy**.

i.e., the fraction of correctly classified samples



(e) River



Forest



Challenge Logistics

1. Team building: 9 teams of 3 people
 - Teams are the same for the exercise sheets and coding challenge
2. Lab on 09 November: coding challenge “mid-term”
 - Check-in/ Q&A
3. Last Kaggle submission + code & presentation submission deadline: **13 December 23:59pm**
4. Final presentation:
 - 14 December during the class + exercises time slot
 - Slide template will be provided
 - Your presence is required!
5. Grading:
 - 20% of course grade

Grading rubric

#	Performance Criteria	Max. Points	Min. Requirements for 2/4 pts	Min. Requirements for 3/6 pts	Min. Requirements for 4/8 pts	Min. Requirements for 5/10 pts
1	Code Submission & Documentation	5	Code for working model is submitted. Valid submission to Kaggle. Code reproduces csv file from Kaggle submission.	Code for working model for both tracks is submitted with minimal documentation (e.g., cells have headers).	Code for working model is submitted with comprehensive documentation (code is wrapped in documented functions/classes). Code and documentation for both tracks are submitted.	Code for working model with excellent documentation (results are visualized e.g., accuracy over training epochs, confusion matrix, etc.) for both tracks.
2	Classification Results (private leaderboard)	10	Better than random guessing performance on test set (Kaggle > 15% accuracy).	TBD during midterm coding challenge check-in	TBD during midterm coding challenge check-in	TBD during midterm coding challenge check-in
6	Final Presentation	5	Presentation is submitted.	Presentation and slides are well structured. Solutions for both tracks are explained.	Presentation and slides are well structured and individual slides are understandable, both tracks are covered.	Presentation and slides are well structured and individual slides are understandable. Provides details and visualizations are convincing (for both tracks).

Tracks

1. Medium models

- Models **up to 10M parameters**
- It is likely your most performant model will be from this track

2. Small models

- Models **up to 1M parameters**
- Also submit predictions from this track on Kaggle to evaluate test accuracy

For both tracks:

- You can implement your own architecture (more details in Lab 04 CNN)
- Or use a pretrained model which you finetune on EuroSAT
- We will check your code for the number of parameters used
- You will have to provide the model weights for which you got your best score, and we will test them

Course Labs and Challenge Roadmap

Event Type	Date	Description	Teaching Assistant	CC Phase
CC 2	9 Nov 2026	Coding Challenge - Mid-Term	Léo/Aron	Coding Phase
Lab 6	16 Nov 2026	Attention	Léo/Aron	Coding Phase
Lab 7	23 Nov 2026	K-Means, EM Clustering	Léo/Aron	Coding Phase
Lab 8	30 Nov 2026	Autoencoder Anomaly Detection	Léo/Aron	Coding Phase
Lab 9	07 Dec 2026	Transfer Learning / Self-Supervised Learning	Léo/Aron	Finalization Phase
CC 3	14 Dec 2026	Coding Challenge – Final Session	Léo/Aron	Finalization Phase

Happy Coding! 🧐



Artificial Intelligence
and Machine Learning

Universität St.Gallen



MCS Machine Learning Class of Fall 2026

Chair of Artificial Intelligence
and Machine Learning [AI:ML]
Rosenbergstrasse 30, CH-9000 St. Gallen
aiml-teaching.ics@unisg.ch

